COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	20/08/26	**DATE SUBMITTED:**	20/08/26
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ACTIVITY NO. 1.1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. What is a search tree in data structures?
A tree based data structure used to store and organize data so that values can be inputted, searched and deleted efficiently.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

Linear search checks each element one by one, while, Binary search repeatedly divides the search in half.

3. What operations / implementations can be performed using binary and linear search operations?

Linear:

Find a specific element, check whether an element exists, find the index of an element, search through unsorted array, and find the first or last occurrence of a value.

Binary:

Find a specific element in a sorted array, find the index of an element, check whether an element exists, find the first or last occurrence of a value, and efficiently search large sorted datasets.

4. What are the advantages in using binary search tree as data structure?

Fast searching, Fast insertion, Fast deletion, Maintains sorted order, Dynamic size, and Efficient data organization.

5. Give an example program using binary search and Linear search.

Search an item using linear search
#include<iostream>

```
using namespace std;
int main(){
    int a[20],i,n,search;
    cout<<"Enter the size:\n";
    cin>>n;
    cout<<"Enter the elements:\n";
    for(i=0;i<n;i++){
        cin>>a[i];
    }
    cout<<"Enter the element to search:\n";
    cin>>search;
    for(i=0;i<n;i++){
        if(a[i]==search){
            cout<<search<<" is found at "<<i+1;
            break;
        }
    }
    if(i==n){
        cout<<search<<" is not present in the array";
    }
    return 0;
}
```

Search an item using binary search

```
#include<iostream>
using namespace std;
int main(){
```

```
int a[20],i,n,low,high,mid,search;
cout<<"Enter the size:\n";
cin>>n;
cout<<"Enter the elements:\n";
for(i=0;i<n;i++){
    cin>>a[i];
}
cout<<"Enter the element to search:\n";
cin>>search;
low=0;
high=n-1;
mid=(low+high)/2;
while(low<=high){
    if(a[mid]==search){
        cout<<"item found at position "<<mid;
        break;
    }
    else if(a[mid]<search){
        low=mid+1;
    }
    else{
        high=mid-1;
    }
    mid=(low+high)/2;
}
if(low>high){
    cout<<"Item not found";
```

```
    }
    return 0;
}
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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